MILITARY SPECIFICATION

# BLOWER, GASOLINE-DRIVEN, AIRCRAFT VENTILATION, PORTABLE, 2,900 CFM, TYPE A-2

This specification is approved for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This specification covers one type of 2,900 cubic feet per minute portable, aircraft ventilation, gasoline-driven blower, designated type A-2.

## 2. APPLICABLE DOCUMENTS

* 2.1 Government documents.

* 2.1.1 Specifications and standards. Unless otherwise specified (see 6.2), the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

| | |
|---|---|
| PPP-B-601 | Boxes, Wood, Cleated Plywood. |

MILITARY

| | |
|---|---|
| MIL-P-116 | Preservation, Method of. |
| MIL-T-704 | Treatment and Painting of Materiel. |
| DOD-D-1000 | Drawing, Engineering and Associated List. |
| MIL-H-7365 | Hose, Airduct, for Ground Heaters. |
| MIL-T-16165 | Interference Shielding, Engine Electrical Systems |

STANDARDS

FEDERAL

| | |
|---|---|
| FED-STD-595 | Colors. |

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: the Engineering Division, San Antonio Air Logistics Center, Kelly AFB, Texas 78241 by using the self addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.



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| | |
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| MIL-STD-105 | Sampling Procedures and Tables for Inspection by Attributes. |
| MIL-STD-129 | Marking for Shipment and Storage. |
| MIL-STD-130 | Identification Marking of U.S. Military Property. |
| MIL-STD-143 | Standards and Specifications, Order of Precedence for the Selection of. |
| MIL-STD-461 | Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference. |
| MIL-STD-808 | Finish, Materials and Processes for Corrosion Prevention and Control in Support Equipment. |
| MIL-STD-810 | Environmental Test Methods and Engineering Guidelines. |
| MIL-STD-831 | Test Reports, Preparation of. |
| MIL-STD-1186 | Cushioning, Anchoring, Bracing,Blocking, and Waterproofing, with Appropriate Test Methods. |
| MIL-STD-1410 | Methods for Selection of Industrial Engines for End Item Application. |

* DRAWINGS

| | |
|---|---|
| 7545352 | Requirement for Finishes, Protective, and Codes for San Antonio ALC Ground and Ground Support Equipment. |

(Copies of specifications and standards required by manufacturers in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

* 2.1.1 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

UNIFORM CLASSIFICATION COMMITTEE

Uniform Freight Classification Rules

(Applications for copies should be addressed to the Uniform Classification Committee, 292 Union Station, 516 West Jackson Boulevard, Chicago, IL 60004.)

AIR MOVING AND CONDITIONING ASSOCIATION

| | |
|---|---|
| AMCA Standard 210-67 | Test Code for Air Moving Devices |

(Application for copies should be addressed to Air Moving and Conditioning Association, 30 West University Drive, Arlington Heights IL 60004.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

| | |
|---|---|
| * ASTM D 3951-82 | Standard Practice for Commercial Packaging. |



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(Application for copies of ASTM publications should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, Pennsylvania 19103.)

2.2 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

* 3.1 First article. This specification makes provisions for first article testing.

3.2 Components. The blower shall consist of the following major components:

| Description | See Requirement |
|---|---|
| a. Frame and fan shroud | 3.7 |
| b. Wheels and wheel lock | 3.8 |
| c. Air duct assembly | 3.9 |
| d. Fan and fan guards | 3.10 |
| e. Gasoline engine | 3.11 |

3.3 Selection of standards and specifications. Standards and specifications for necessary commodities and services not specified herein shall be selected in accordance with MIL-STD-143 except as provided in 3.3.1 and 3.3.2.

3.3.1 Commercial parts. Commercial parts having suitable properties may be used where, on the date of invitation for bids, there are no suitable standard parts. In any case, commercial utility parts, such as screws, bolts, nuts, and cotter pins, having suitable properties may be used, provided:

a. They can be replaced by the standard parts (MS or AN) without alteration.

b. The corresponding standard part numbers are referenced in the parts list and, if practical, on the contractor's drawings.

3.3.2 Standard parts. With the exception in 3.3.1, MS and AN standard parts shall be used where they suit the purpose. They shall be identified on the drawings by their part numbers.

3.4 Material. It is the intent of this specification that the quality of the material used in the construction of equipment be at least equal to that covered by referenced specifications. Where material is not in accordance with the requirements of referenced specifications, it must be clearly demonstrated that an improvement in operating characteristics or a saving in weight without sacrifice in reliability can be accomplished thereby, and that such substitutions do not preclude the subsequent use of specification material in effecting repairs or replacements. Specific approval by the bureau or agency concerned under the individual contract or order shall be obtained for use of materials not in accordance with the requirements of referenced specifications. The manufacturer's request for approval of such use shall be accomplished by shop drawings or plans clearly illustrating the use of the material.



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3.4.1 Protective treatment. When materials are used in the construction of the blower that are subject to deterioration when exposed to climatic and environmental conditions likely to occur during service usage, they shall be protected against such deterioration in a manner that will in no way prevent compliance with the performance requirements of this specification. The use of any protective coating that will crack, chip, or scale with age or extremes of climatic and environmental conditions shall be avoided.

3.5 Design and construction. The blower shall be designed and constructed to operate continuously for a period of 200 hours at rated capacity on any gasoline, including all leaded types, and intermittently for a period of 750 hours when receiving normal servicing, that is, valve grinding and engine-head cleaning.

3.6 Performance. The blower shall satisfactorily perform under all conditions specified herein. The blower assembly shall have a minimum air delivery of 2,900 cubic feet per minute with the duct assembly attached, when operated at a speed of 3,600 revolutions per minute and at standard atmospheric pressure.

3.6.1 Environmental. The blower shall withstand exposure to and function properly under the following environmental conditions as described under 4.4:

a. High temperature

(1) Operation 140° Fahrenheit (F)
(2) Exposure 160°F

b. Humidity from 0 to 100 percent

c. Salt fog

d. Sand and dust

3.7 Frame and fan shroud. The blower frame and fan shroud shall be constructed of light-gage sheet steel perforated where practical to reduce the weight of the unit to a minimum. Aluminum may be used provided it is of sufficient strength and thickness to contain the fan blade if the blade becomes loose or fails.

3.7.1 Handle. A handle attached to the frame shall be provided. The handle shall be made to collapse within the dimensions of the unit without the use of tools. The handle shall be reinforced at the point of attachment to the frame to insure against failure due to operating vibration.

3.7.2 Supporting brackets. All support brackets, supporting component members of the frame of the unit or to the other component parts, shall be adequately reinforced to insure against failure due to operating vibrations.

3.8 Wheels and wheel lock. The blower shall be mounted on two rubber wheels to cushion the vibration of the gasoline engine and to provide easy portability. A leg support pivoted to provide an automatic wheel lock when the unit is set in the operating position shall be provided.



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3.8.1 Wheels. The wheels shall contain semi-pneumatic (hollow center) rubber tires, 6 inches in diameter by 1.75 or 2 inches wide.

3.8.2 Wheel lock. The wheel lock, together with the rubber cushions on the leg support, shall prevent the unit from creeping under engine vibrations and fan reaction.

3.8.2.1 The wheel lock shall be attached to the frame by pivot members of adequate size to insure against failure in operation. The ends of the wheel lock shall be joined in a manner to prevent spreading during operation, and shall be capable of being disjoined, if necessary, to permit replacement of the rubber cushioning collars. Rubber or composition collars having a high degree of resistance to abrasion shall be provided on the supporting section of the wheel lock.

3.9 Air duct. The air duct shall be 14 inches in diameter and 20 feet long; it shall have four straps, each 36 inches long with buckles, attached to the discharge end for convenience in tying the air duct in service position. The other end of the air duct shall be securely attached to the fan end of the air and storage compartment.

3.9.1 The air duct material shall conform to that specified in MIL-H-7365. Color shall be No. 24042 per FED-STD-595.

3.9.1.1 The air duct shall be formed from a strip of duck wound helically and having overlapping edges with a partially contracted spring wire between and stitched with a two-needle sewing maching. The free length pitch of the spring wire shall be substantially greater than its pitch in the finished seam to exert axial tension on and prevent transverse collapse of the air duct. The duct shall incorporate a high abrasion-resistant scuff strip of polyvinylchloride material, or equal, over the helical-type stiffening wire in such manner that the basic fabric will be protected from excessive abrasion when dragged over level surfaces.

3.9.1.2 The air duct shall be securely attached at one end to the air duct storage compartment.

3.9.1.3 The air duct shall be capable of collapsing in bellows fashion into the air duct storage compartment.

3.9.1.4 The air duct shall be provided with a helical-type stiffening wire with an 8-inch finished seam pitch.

3.9.2 Air duct storage compartment. An air duct storage compartment 15.75 inches in diameter and 16 inches long shall be provided for the air duct. It shall be fabricated of aluminum alloy sheet and shall be provided with a cover to make the compartment weathertight, except the fan shroud opening. The cover shall be attached by a chain to prevent loss.

3.9.2.1 The air duct storage compartment shall be permanently fastened to a 20 degree elbow. The 20 degree elbow shall be fastened to the discharge of the fan shroud.

3.9.3 The assembly shall be capable of being attached to the shroud in four positions to direct the air duct 20 degrees up, down, right, or left.



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3.9.4 The air duct, and air duct storage compartment assembly shall be capable of being attached to the fan shroud without the use of tools.

3.9.5 The air duct and air-duct-storage-compartment assembly shall have no loose parts.

3.10 Fan and fan guards. The fan shall be a 14-inch diameter, axial flow propeller type, capable of a minimum air delivery of 2,900 cubic feet per minute through the duct specified herein, at fan speeds ranging from 3,400 to 3,600 revolutions per minute.

3.10.1 Fabricated wire guards shall be installed to guard both the inlet and the outlet. Openings shall be no larger than 0.50 inches.

* 3.11 Gasoline engine. The gasoline engine shall conform to MIL-STD-1410, Group I, four stroke cycle, Class I or II and selected to withstand all tests and conditions specified herein. The engine selected shall be the "heavy duty" type containing a cast front cylinder block or bore liner, steel connecting rod and ball or roller crank shaft main bearings.

3.11.1 Governor. The governor shall be of a type which can be set to govern the engine at any speed between idle and the speed required for the fan to deliver air under the rated conditions specified in 3.6.

3.11.2 Manual throttle. A manual throttle to control the speed from full-governed speed to full-idling speed shall be provided the gasoline engine.

3.11.3 Air cleaner. An air cleaner shall be installed in the air-intake line of the gasoline engine to insure a clean supply of air to the carburetor.

3.11.4 Magneto ignition system. A magneto ignition system shall be provided for the gasoline engine. Complete shielding against radio interference shall be provided in accordance with MIL-I-16165 when the blower is tested to the requirements of MIL-STD-461, class IIIC.

3.11.5 Starting cord. A starting cord for manually starting the gasoline engine shall be provided.

3.11.6 Gasoline tank. A gasoline tank shall be provided. The gasoline tank shall have sufficient fuel capacity to insure a minimum of 2 hours of operations at full load speed. The gasoline tank shall be attached to the supporting bracket by vibration-proof fastening.

3.11.7 Exhaust line. A flexible exhaust line made of lightweight, heat-resisting alloy tubing covered with woven asbestos or equivalent tubing, shall be provided.

3.11.7.1 The exhaust line shall be not less than 6 nor more than 7.5 feet long with a muffler and handle attached at the discharge end. The muffler shall be the type that readily permits removal of accumulated deposits.

3.11.7.2 A means shall be provided for stowing the exhaust line on the blower when the unit is not operating.



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3.11.8 Stop switch. A stop switch shall be provided. The stop switch shall be installed so that vibration will not cause displacement of any of its parts that may short out the magneto and stop the engine.

3.11.9 Fuel flow adjustment. A carburetor fuel flow adjustment shall be provided and shall contain a locknut to maintain the setting when subjected to operating vibration.

* 3.12 Interchangeability. All parts having the same manufacturer's part number shall be functionally and dimensionally interchangeable. The drawing number requirements of DOD-D-1000 shall govern changes in the manufacturer's part numbers.

3.13 Dimensions. The overall dimensions of the blower without the air duct attached and with the handle down, shall not exceed 24.625 inches high, 21 inches wide and 20.50 inches long. With the duct assembly attached and with the duct collapsed in the stowed position, the overall dimensions shall not exceed 24.625 inches high, 21 inches wide, and 36.75 inches long.

3.14 Weight.

* 3.14.1 The maximum weight of the complete blower, excluding gasoline or oil, but including the air duct, shall not exceed 110 pounds.

* 3.14.2 The weight of the blower excluding gasoline, oil, and air duct, shall not exceed 82 pounds.

* 3.15 Finish. All parts of the blower (exluding the air duct and engine) shall be finished in accordance with MIL-STD-808 in conjunction with AF drawing 7545352, except the fan blade shall be color 11136. Treatment and painting of the engine shall be in accordance with the manufacturers standard commercial practice.

3.16 Instruction plate. An instruction plate, permanently and legibly marked with precautions and with the steps for starting and stopping the blower, shall be mounted on the blower in full view of the operator.

3.17 Identification of product. Equipment, assemblies, and parts shall be marked for identification in accordance with MIL-STD-130.

3.18 Workmanship.

3.18.1 General. The blower, including all parts and accessories, shall be fabricated and finished in a thoroughly workmanlike manner. Particular attention shall be given to freedom from blemishes, defects, burrs, and sharp edges; accuracy of dimensions, marking of parts and assemblies; thoroughness of welding, brazing, painting, wiring, and riveting; alignment of parts and tightness of assembly screws, bolts, et cetera.

3.18.2 Cleaning. The blower shall be thoroughly cleaned, and metal chips, and other foreign material removed during and after final assembly.



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4. QUALITY ASSURANCE PROVISIONS

* 4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the contractor may use his own or any other facilities suitable for the performance of the inspection requiremnts specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

* 4.2 Classification of tests. The inspection and testing of blowers shall be classified as follows:

a. First article tests.
b. Quality conformance inspection.
c. Acceptance tests.

* 4.2.1 First article tests. First article tests shall consist of all tests described under 4.4. Acceptable delivery of production items on a Government contract within the preceding two years shall be grounds for waiving first article testing.

* 4.2.2 First article test sample tested by the contractor. The contractor shall subject one blower to all the tests specified in 4.4. If the unit fails during first article testing and repairs can be made to the unit without any internal unit disassembly, the testing may commence again at the point of failure. A failure of this nature will not constitute a first article failure. However, anomalies or failures of this nature and corrective actions taken will be noted in the first article test report.

* 4.2.3 First article test report. After the contractor completes the first article tests, he shall prepare a first article test report in accordance with MIL-STD-831 and furnish three complete copies of the report to the procuring activity.

4.3 Quality conformance inspection. Quality conformance inspection shall consist of a sampling plan and tests.

4.3.1 Sampling plan and tests.

* 4.3.1.1 Lot. One blower shall be selected at random (by the QAR) from every 50 or fraction thereof on the contract or order and subjected to examination of product and capacity tests.

4.3.1.2 Rejection and retest. When one or more blowers from a lot fail to meet the specification, acceptance of all blowers in the lot will be withheld until the extent and cause of failure are determined. After corrections have been made, all necessary tests shall be repeated.

4.3.1.2.1 Individual tests may continue. For production reasons, individual tests may be continued pending the investigation of a sampling test failure. Final acceptance of the entire lot shall not be made until it is determined that the lot meets all the requirements of this specification.



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4.3.2 Defects in items already accepted. The investigation of a test failure could indicate that defects may exist in blowers already accepted. If so, the contractor shall fully advise the procuring activity of all defects likely to be found and methods of correcting them.

4.4 Test methods.

4.4.1 Capacity test. The cubic feet of air delivery shall be measured in accordance with AMCA Standard 210-67 (see 3.6).

4.4.1.1 Acceptance test. Each blower submitted for acceptance shall be subjected to the examination of product test.

4.4.2 Endurance tests. The blower shall be operated continuously for 200 hours at rated capacity. At the end of the 100 hours, the blower shall be checked for compliance with all the requirements specified herein.

4.4.3 Environmental tests. The blower shall be tested to determine its starting and operating performance characteristics at any temperature from 140°F ambient temperature down to and including 100°F. The blower shall be soaked at each testing temperature for 24 hours before the starting and operating tests are performed at each testing temperature.

4.4.4 High temperature tests. The blower shall be tested in accordance with procedure I of MIL-STD-810.

4.4.5 Humidity. The blower shall be tested in accordance with procedure I of MIL-STD-810.

4.4.6 Salt fog. The blower shall be tested in accordance with procedure I of MIL-STD-810.

4.4.7 Dust. The blower shall be tested in accordance with procedure I of MIL-STD-810.

4.4.8 Electromagnetic interference test. The blower shall be tested in accordance with class IIIC of MIL-STD-461.

* 4.5 Examination of product test. The blower shall be inspected to determine compliance with the requirements specified herein. The inspection is to include parts, materials, workmanship, dimensions and markings. A check list shall be prepared.

4.6 Inspection of the preservation, packaging, packing and marking for shipment and storage. Sample items or packs and the inspection of the preservation, packaging, packing and marking for shipment and storage shall be in accordance with the requirements of Section 5, or the documents specified therein.

* 5. PACKING

5.1 Preservation. Preservation shall be level A, or Commercial, as specified (see 6.2).

5.1.1 Level A.

5.1.1.1 Cleaning and drying. The blower shall be cleaned in accordance with process C-1 and dried in accordance with procedures D-1 and D-4 of MIL-P-116.



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5.1.1.2 Unit packaging. The blower shall be preserved and packaged in accordance with MIL-P-116, Method II.

5.1.2 Commercial. Blowers shall be preserved in accordance with ASTM D 3951.

5.2 Packing. Packing shall be level A, B, or Commercial as specified (see 6.2).

5.2.1 Level A. The blower, preserved and packaged as specified in 5.1.1, shall be packed in a shipping container conforming to PPP-B-601, domestic type, style B. Containers shall be of uniform size and shape and of minimum tare and cube.

5.2.2 Level B. The blower, preserved and packaged as specified in 5.1.1, shall be packed in a shipping container conforming to PPP-B-601, overseas type, style B. Containers shall be of uniform size and shape and of minimum tare and cube.

5.2.3 Commercial. Blowers preserved as specified on 5.1, shall be packed in accordance with ASTM D 3951-82.

5.3 Physical protection. Interior cushioning, anchoring, blocking and bracing shall be in accordance with MIL-STD-1186.

* 5.4 Marking. In addition to any special marking required by the contract or order, containers shall be marked in accordance with MIL-STD-129 or ASTM D 3951-82 as applicable.

6. NOTES

6.1 Intended use. The type A-2 blower covered by this specification is intended primarily for use in cooling airplane fuselages during maintenance periods in maintenance shelters, portable hangers, and other similar inclosures.

6.2 Ordering data.

6.2.1 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification.

b. Selection of applicable levels of preservation, packaging and packing (see 5.1 and 5.2).

6.3 The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.



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Custodians:
Air Force - 99
Army - AV
Navy - AS

Preparing activity:
Air Force - 82

Project 1730-0273



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# STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

*(See Instructions -- Reverse Side)*

| 1. DOCUMENT NUMBER | 2. DOCUMENT TITLE |
|---|---|
| MIL-B-3402C | BLOWER, GASOLINE-DRIVEN, AIRCRAFT VENTILATION, PORTABLE, 2,900 CFM, TYPE A-2 |

| 3a. NAME OF SUBMITTING ORGANIZATION | 4. TYPE OF ORGANIZATION *(Mark one)* |
|---|---|
| | ☐ VENDOR |
| b. ADDRESS *(Street, City, State, ZIP Code)* | ☐ USER |
| | ☐ MANUFACTURER |
| | ☐ OTHER *(Specify)*: |

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

| 7a. NAME OF SUBMITTER *(Last, First, MI)* — Optional | b. WORK TELEPHONE NUMBER *(Include Area Code)* — Optional |
|---|---|
| c. MAILING ADDRESS *(Street, City, State, ZIP Code)* — Optional | 8. DATE OF SUBMISSION *(YYMMDD)* |

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